

Madame

TD 6 

Exo 4:

$$\begin{aligned}
 1. \quad \max z_1 &= 18x_1 + 12x_2 + 17x_3 + 2x_4 & (1, 1, 0, 0) &\rightarrow (30, 14) \\
 \max z_2 &= 3x_1 + 11x_2 + 7x_3 + 15x_4 & (1, 0, 1, 0) &\rightarrow (35, 10) \Rightarrow x_1^* \\
 \text{s.t. } \begin{cases} 4x_1 + 5x_2 + 6x_3 + 5x_4 \leq 10 \\ x_i \in \{0, 1\}, i = 1, \dots, 4 \end{cases} & & (1, 0, 0, 1) &\rightarrow (20, 18) \\
 & & (0, 1, 0, 1) &\rightarrow (14, 26) \Rightarrow x_2^*
 \end{aligned}$$

On optimise chaque critère seulement !

Proouvons le pt° idéal et nadir.

$$\begin{aligned}
 x_1^* &= (1, 0, 1, 0) \Rightarrow \begin{pmatrix} 35 & 10 \\ 14 & 26 \end{pmatrix} \\
 x_2^* &= (0, 1, 0, 1)
 \end{aligned}$$

$$\begin{aligned}
 \hookrightarrow y^0 &= (35, 26) \\
 y^N &= (14, 10)
 \end{aligned}$$

$$2. \quad \min \max \left\{ w_1 \frac{35 - z_1}{35 - 14}, w_2 \frac{26 - z_2}{26 - 10} \right\} + \varepsilon \left(\frac{35 - z_1}{35 - 14} + \frac{26 - z_2}{26 - 10} \right)$$

$$\begin{cases} z_1 = 18x_1 + 12x_2 + 17x_3 + 2x_4 \\ z_2 = 3x_1 + 11x_2 + 7x_3 + 15x_4 \\ 4x_1 + 5x_2 + 6x_3 + 5x_4 \leq 10 \\ x_i \in \{0, 1\} \quad i = 1, \dots, 4 \end{cases}$$

$$\begin{aligned}
 \Rightarrow \quad \min z &+ \varepsilon \left(\frac{35 - z_1}{35 - 14} + \frac{26 - z_2}{26 - 10} \right) \\
 \begin{cases} z &\geq w_1 \left(\frac{35 - z_1}{21} \right) \\ z &\geq w_2 \left(\frac{26 - z_2}{16} \right) \\ z_1 &= 18x_1 + 12x_2 + 17x_3 + 2x_4 \\ z_2 &= 3x_1 + 11x_2 + 7x_3 + 15x_4 \\ 4x_1 + 5x_2 + 6x_3 + 5x_4 &\leq 10 \\ x_i &\in \{0, 1\} \quad i = 1, \dots, 4 \end{cases} \quad \Rightarrow \quad \begin{aligned} x^* &= (1, 0, 0, 1) \\ y^* &= (20, 18) \end{aligned}
 \end{aligned}$$

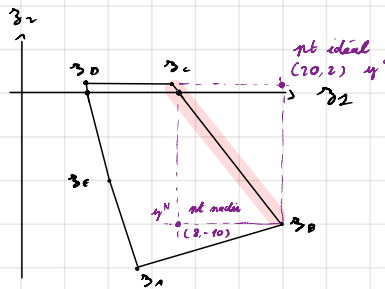
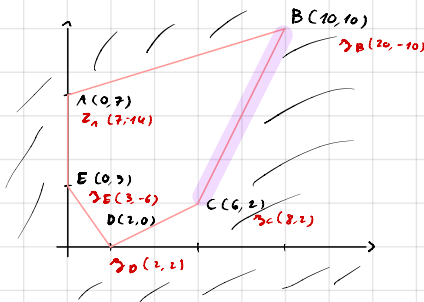
Exo 2:

1/2/3. $\max z_2 = x + 4y$

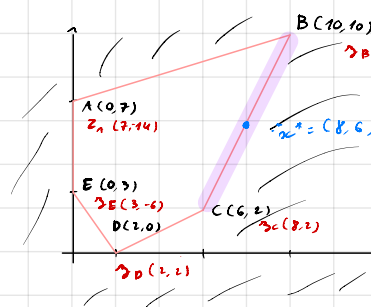
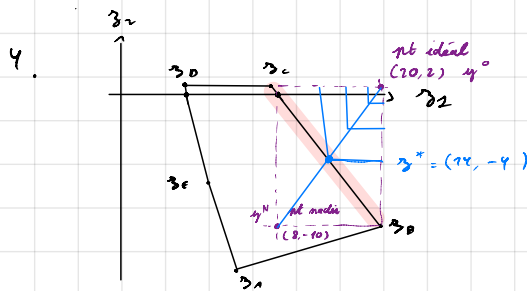
(On met tout ces max (ou min))

$\max z_2 = +x - 2y$

Espace de sol°



$$\begin{array}{r} 20 \\ 8 \end{array} \begin{array}{r} -10 \\ 2 \end{array}$$

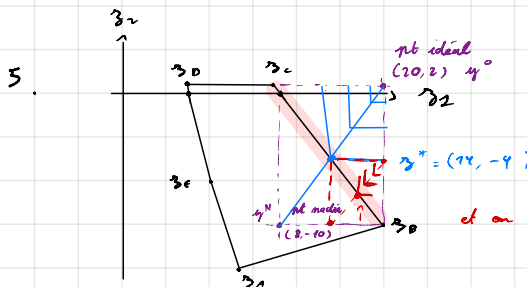


$$\begin{cases} x + 4y = 14 \\ x - 2y = -4 \\ 3y = 18 \end{cases}$$

$$x = 8$$

$$y = 6$$

$$\begin{array}{r} 20 \\ 8 \end{array} \begin{array}{r} -10 \\ 2 \end{array} \begin{array}{r} 14 \\ -4 \\ 18 \end{array}$$



et on recommence, de même pour q6 mais de l'autre sens.

